

44% ticket deflection without a single hallucinated answer

Coilforge — developer-tooling SaaS, ~210 customers · Engagement Q4 2025

Industry	B2B developer tooling — distributed-systems testing platform
Service line	AI Customer Support Agent
Engagement length	7-week build, 10 months managed (ongoing)
Investment	€7,900 build + €580/month managed
Stack	Front, Postgres + pgvector, in-app widget, Slack Connect

The Challenge

Coilforge's two-person support team was buried. The product is technical — engineers use it to inject controlled failures into distributed systems — and customer questions ranged from “how do I install the CLI” to “why does the fault-injection sidecar drop connections under load.” Median response time was 5.1 hours. The CTO, Adi Patel, told us in our first call: “If you give me a chatbot that hallucinates a single API parameter, I will rip it out the next morning. Our customers will trust it once and never again.”

The team had previously trialled an off-the-shelf support chatbot from a well-known vendor. It had answered six questions wrong in its first week, including telling a customer that a command-line flag existed when it did not. They removed it on day eight.

Our Approach

Citation-grounded retrieval was the non-negotiable architectural choice. Every reply the agent produces includes inline links to the documentation chunks that support it. If the retrieval step does not return at least one chunk above a confidence threshold of 0.78, the agent does not generate an answer — it explicitly tells the user it does not have the information and routes to a human. There is no fallback to general LLM knowledge, ever.

We indexed Coilforge's documentation site, the public changelog, and 18 months of resolved support tickets into a pgvector store on their existing Postgres. Embeddings refresh nightly. Tier-2 questions (ambiguous, partially-covered) are not auto-answered — the agent drafts a reply that the human reviewer can approve, edit, or reject from inside Front. Tier-3 questions (billing, contracts, enterprise procurement) bypass the agent entirely.

For enterprise customers using Slack Connect, we deployed a scoped-down version of the same agent that only answers from documentation and explicitly defers to the customer's named CSM for everything else. This was the part of the build that took the longest to get right, because Slack threading semantics make tier-1/tier-2 boundaries less obvious than in email.

Results

44%	0	18 min	+11
tier-1 deflection by month two	hallucinated answers in 10 months	median first response (was 5.1h)	CSAT improvement

The deflection number is the headline, but the more important result is the zero-hallucination figure across roughly 6,400 agent-handled conversations. We define hallucination strictly: any reply where the agent claims a behaviour, parameter, or feature that does not exist in the documentation or codebase. Coilforge audits a sample of 50 conversations per week, and the running count remains zero. The architectural choice to refuse rather than generate when retrieval is weak is the single biggest reason.

In the client's words

"I came in expecting to remove this within a quarter, the way we removed the last one. Instead, I find myself defending it in our weekly all-hands when someone asks why we still need two support engineers. The answer is that we do, but they spend their days on the hard problems, not on telling people which version of the CLI to download."

Adi Patel, CTO — Coilforge

Lessons Learned

- In technical-product support, an agent that refuses to answer is worth more than an agent that answers confidently and wrongly.
- Retrieval-confidence thresholds must be tuned per corpus. Coilforge's threshold of 0.78 was the result of two weeks of evaluation, not a default.
- Tier-2 'draft for human review' is the underrated win. It compounds support team productivity without any of the trust risk of auto-send.
- Slack Connect support agents are a different product than email or in-app — threading and synchronous expectations change the interaction model.